

Program 7

Aim:- Implement of Component diagram for LMS

Component Diagram:- A component diagram is a powerful tool in UML (Unified modeling language) for visualizing the structure & organization of a software system.

Description:- A component Diagram shows the structural relationship between the components of a software system. It helps model the physical aspects of the system, as files, libraries, executables & packages & how they interact to form the overall application architecture.

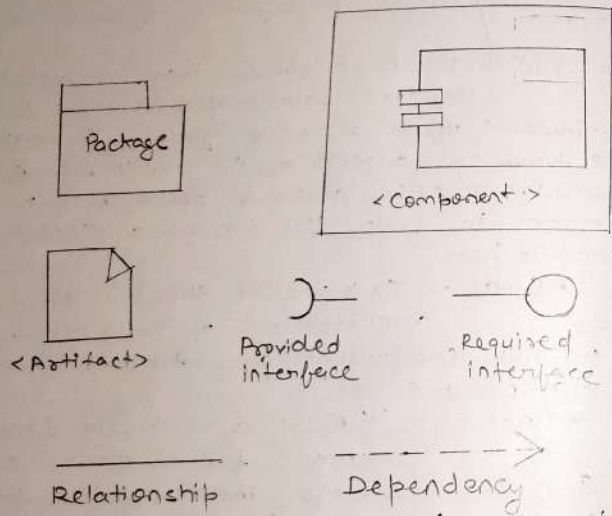
- Goal - To visualize the component based architecture of a system, focusing on how components are ~~wired~~ wired together via interfaces.

- Analogy - Think of a component Diagram as a schematic for an electronic device.

It shows the major integrated circuits (components) the plugs (interface) & how the chips are connected to one another on the board.

Importance:- Component diagrams are vital for

- Understanding System Architecture, Provide high level view of structure.



- Clearly identifies reusable parts of system.
- Dependencies are clearly mapped to easy to maintain & upgrade.
- Form basic deployment diagram.
- Helps bridge the gap b/w logical design & physical implementation.

Notation:-

- Component → interaction & communication occur via interfaces.
- Package → combines multiple elements in a system
ex. class, components, interface into a group.
- Artifacts → physical unit of information
- Provided interfaces → one / more clearly defined interfaces that provide functions
- Required interface → required interface that receives functions, services or data from outside.
- Relationship → Connectors b/w components.
- Dependency → Indicates a dependency b/w two parts of a system.

- Steps
- (1) Identify the system scope & requirements.
 - (2) Identify & define components.
 - (3) Identify provided & required interfaces.
 - (4) Identify relationship & dependencies.
 - (5) Identify nodes.
 - (6) Draw diagram.

Viva QuestionQ1
Ans

What is the purpose of a Component diagram?
The primary purpose of a components diagram is to show the organization & dependencies among software components. It models the static implementation view of a system, illustrating the physical structure of the code, such as files, libraries and executables, & how they interrelate to form the larger system.

Q2
Ans

What is a 'Component' in a UML diagram?
A component is defined as a modular, significant & replaceable part of the system that packages implementation & exposes a set of well-defined interfaces.

Q3
Ans

What are the key element of a component diagram?

The key elements are components (modular parts of the systems); interfaces (collection of operation a components offer or requires), ports (interaction points on a component's boundary) & assembly connector (which link components & their interfaces)

Q4
Ans

How are interfaces shown in a components diagram?
Interfaces can be shown using a 'lollipop' symbol for provided interfaces (operation a component offers ()), an open half-cycle symbol for required interfaces ().

Teacher's Signature _____

Q5 What is an assembly connector?

Ans An assembly connector is used to connect components, showing how a provided interfaces of one component is connected to required interface of another component. It visually represents the dependency between the components.